

PRODUCT REQUIREMENTS DOCUMENT

Designer Blouse AI Platform

V1 Catalog, AI Studio & WhatsApp Order Flow

Document Version	1.1.0
Target Market	India — Mobile-first
Status	Approved for Implementation
V1 Commercial Model	Assisted WhatsApp ordering

This document defines the V1 product, functional, technical, operational, and launch requirements. Implementation code and source-code examples are intentionally excluded.

1. Executive Summary & Scope

The Designer Blouse AI Platform is an India-focused, mobile-first designer blouse fabric catalog combining product discovery, saree-to-blouse color matching, and an AI Blouse Studio. Customers can browse curated unstitched blouse materials, upload a saree or fabric image, sample colors with an interactive eyedropper, discover visually compatible fabrics, and generate three photorealistic blouse views: front, back, and 3/4.

V1 is deliberately optimized for market validation and low operational overhead. It uses an assisted WhatsApp Business ordering flow instead of automated checkout, payments, inventory reservation, or shipping. The architecture must preserve clear feature boundaries so full commerce capabilities can be enabled in V2 without replacing core catalog and studio components.

1.1 V1 In Scope

- Mobile-first responsive catalog and product detail pages.
- Saree photo upload with interactive eyedropper and CIELAB / CIE76 color matching.
- Admin color tagging using Hex values with automatic CIELAB calculation.
- AI Blouse Studio with fabric upload and structured style selection.
- Three-view photorealistic generation: front, back, and 3/4.
- Persistent public design permalinks using /d/:id.
- Dynamic WhatsApp Business enquiry redirection.
- Free-tier-oriented infrastructure using Next.js, Supabase, Cloudflare R2, and an asynchronous job layer.

1.2 V2 / Feature-Flagged Scope

- Automated cart, checkout, and coupons.
- Razorpay or Cashfree domestic payment integration.
- Atomic length-based inventory reservations.
- Shiprocket courier integration and automated AWB tracking.
- Interactive 3D mesh viewer or real-time 360° rotation.

1.3 Product Goals

- Validate demand for AI-assisted designer blouse visualization.
- Reduce friction from saree color selection to relevant fabric discovery.
- Convert high-intent users into WhatsApp enquiries.
- Keep V1 infrastructure and operational cost low.
- Create a clean technical foundation for V2 commerce activation.

2. User Journeys & Workflow Specifications

2.1 Customer Discovery & Color Match

Step	User Action	System Behaviour
1	Browse catalog	User browses and filters fabrics by fabric type, occasion, and price.
2	Match My Saree Color	User uploads a saree image through file picker or camera capture.
3	Sample shade	Interactive loupe / eyedropper lets the user sample the saree body or border.

Step	User Action	System Behaviour
4	Fine-tune	Lightness and saturation controls allow correction for indoor lighting or flash glare.
5	Match	Sampled color is converted to CIELAB and compared against available catalog fabrics.
6	Review	Results are ordered by perceptual color distance and presented in a product grid.
7	Visualize	User launches AI Blouse Studio from a product page or direct upload.
8	Customize	User selects neckline, sleeve cut, back styling, and occasion.
9	Generate	Asynchronous processing creates front, back, and 3/4 photorealistic views.
10	Enquire	User selects Order via WhatsApp to continue sizing, availability, pricing, and payment manually.

2.2 Admin Workflow

- **Product creation:** Admin enters SKU, title, fabric type, price per unit, unit type, description, availability, and product imagery.
- **Color calibration:** Admin samples the primary fabric color using an on-screen eyedropper. The selected Hex value is stored and converted automatically to CIELAB coordinates.
- **AI generation audit:** Admin reviews generation volume, failures, latency, and popular design styles.

3. Technical Architecture & Stack

Layer	Technology	Purpose / Selection Rationale	V1 Cost Target
Frontend & SSR	Next.js 14+ / App Router / TypeScript / Tailwind CSS	SEO-friendly server rendering, unified application layer, and mobile-first performance.	■0 target
Database & Auth	Supabase / PostgreSQL / Auth	Relational product data, authentication, and database-backed color matching.	■0 target
Object Storage & CDN	Cloudflare R2 + CDN	Fabric uploads, catalog imagery, and generated outputs with low-cost storage and egress.	■0 target
Queue & Worker	Upstash Redis + QStash	Decouples web requests from long-running AI generation tasks and supports asynchronous callbacks.	■0 target
AI Generation	Fal.ai or Replicate	Photorealistic blouse generation with reference-image conditioning and multi-view consistency.	Usage based
Messaging	WhatsApp Click-to-Chat	Manual lead handoff without a V1 BSP or checkout integration.	■0 target
Domain & DNS	Cloudflare Registrar / DNS	Domain management, DNS, SSL/TLS, and basic edge protection.	~■100/month amortized

3.1 Architectural Principles

- Mobile-first UX with responsive layouts optimized for approximately 360–430 px widths.
- Asynchronous AI processing so generation does not depend on a long-lived browser request.
- Object storage for user uploads and generated assets; database stores metadata and references.
- Feature-flagged commerce boundaries so V1 WhatsApp conversion can be replaced by V2 checkout.
- Public design pages must remain independently addressable and shareable.

4. Perceptual Color Matching Engine

The platform shall use CIELAB rather than raw RGB or Hex distance for catalog matching because CIELAB is better aligned with perceived color difference. Customer and admin colors are transformed from sampled sRGB values through CIEXYZ using the D65 standard illuminant and then into CIELAB coordinates.

4.1 Matching Metric

V1 uses CIE76 color difference (ΔE), calculated as the Euclidean distance between two CIELAB coordinates. The implementation shall return the closest available fabrics first.

ΔE Range	Interpretation	V1 Behaviour
≤ 3.5	High-accuracy match	Prioritize as an exact or near-exact tone match.
> 3.5 and ≤ 8.0	Acceptable tonal match	Include as complementary / harmonious results.
> 8.0	Low similarity	Exclude from primary match results.

4.2 Color-Match Functional Requirements

- Customer uploads a saree image from file picker or camera capture.
- The image preview supports an interactive magnifying loupe at approximately 2.5× zoom with a central crosshair.
- The selected pixel is sampled from the rendered image and converted to a Hex value.
- Lightness and saturation controls support adjustment of approximately $\pm 20\%$ to compensate for lighting conditions.
- The adjusted color is converted to CIELAB and dispatched to the catalog matching service.
- Results display closest matches first and expose the relevant product imagery and price.

5. AI Blouse Studio

5.1 Input Requirements

- Accepted formats: JPG, PNG, and WebP.
- Maximum upload size: 10 MB.
- Minimum image dimensions: 800 × 800 px.
- The interface must reject unsupported formats, oversized files, and images below the minimum dimensions.

5.2 Style Taxonomy

Attribute	Supported V1 Values
Neckline	Round; V-Neck; Deep V; Square; Sweetheart; Boat Neck; Halter; High Neck
Sleeves	Sleeveless; Short; Elbow-Length; Full Sleeve; Puff Sleeve; Bell Sleeve
Back	Solid Closed; Deep Back; Keyhole; Tie-Back / Dori; Cut-Out Designer
Occasion	Minimalist Everyday; Festive Silk; Contemporary Evening; Bridal Zari

5.3 Generation Requirements

- The generation request must preserve the uploaded fabric's pattern, embroidery, and target color as closely as practical.
- The target Hex shade must be included as a hard design constraint in the generation specification.
- Negative constraints must discourage color shift, altered embroidery, faded fabric, and distorted seams.

- The initial front view establishes the visual identity for subsequent back and 3/4 views.
- The final result is a coherent set of three photorealistic views of the same blouse design.
- Generation runs asynchronously and must provide clear pending, processing, completed, and failed states.

5.4 Usage Quota

Each authenticated user may generate a maximum of 3 complete designs in a rolling 24-hour cycle. Each complete design represents three image operations, for a maximum of 9 image operations per user per cycle.

6. Data Model & Persistence Requirements

The V1 database shall maintain normalized product and user profile records, generated-design records, and AI usage/audit records. The following requirements define the logical schema; implementation details such as SQL migration syntax are intentionally excluded from this PRD.

Entity	Required Data	Key Behaviour
User Profile	User ID, email, full name, phone number, created timestamp	Linked to authenticated users; deletable without orphaning required design history.
Product	SKU, title, description, fabric type, price, unit type, imagery, Hex colors, color family, CIELAB values, availability	Unique SKU; searchable by availability and perceptual color distance.
AI Design	User, product, source fabric, target Hex, style attributes, three image URLs, status, error, WhatsApp-click flag, timestamp	Publicly addressable through design ID after completion.
AI Usage Log	User, design, provider, provider job ID, operation count, latency, status, timestamp	Supports quota enforcement, operational monitoring, and provider audits.

7. WhatsApp Enquiry & Public Design Flow

7.1 Order via WhatsApp

- The primary conversion CTA shall be displayed on product detail pages and completed AI Studio results.
- The system shall generate a URL-encoded WhatsApp Click-to-Chat link for the configured business number.
- The enquiry message shall include product name and SKU, price, matched color and ΔE where applicable, selected blouse style, and the public design permalink.
- The message shall request manual confirmation of availability and tailoring timeline.
- V1 does not process payment, finalize stock reservations, or issue shipping labels through the web application.

7.2 Public Design Permalink

Each completed design shall have a public route in the form /d/:id and shall not require authentication to view.

- Uploaded fabric reference.
- Front, back, and 3/4 AI views with image zoom.
- Selected style specifications.
- Target Hex color swatch.
- Sticky secondary CTA: Chat with Stylist on WhatsApp.

8. Feature-Flag Strategy

V1 commerce functionality shall remain disabled by default and be isolated behind environment-driven feature flags. The flag strategy must prevent cart, payment, and shipping UI from appearing when the corresponding capabilities are disabled.

Feature Area	V1 Default	V2 Purpose
Commerce	Disabled	Enables cart and commerce actions.
Payments	Disabled	Enables Razorpay / Cashfree payment flow.
Shipping	Disabled	Enables direct courier and tracking integration.

9. Financial Budget & Operational Model

9.1 Launch / Development Cost

Cost Item	Estimated Cost
Domain registration	■1,100 annual payment
AI coding tools	■5,000–■7,500
AI model quality pilot / test generation	■1,500
Total estimated launch cost	■7,600–■10,100

9.2 Monthly Operating Model

Component	V1 Estimate
Frontend hosting	■0 target
Database & authentication	■0 target
Object storage & egress	■0 target
Background job queue	■0 target
AI image generation	Approx. ■1,250/month at 500 operations
Domain amortization	Approx. ■100/month
Total estimated ongoing cost	Approx. ■1,350/month

Planning assumption: The estimate is based on approximately 100 completed designs per month and approximately 5 AI operations per design when retries are included. Actual AI spend will vary by provider, model, image resolution, and retry rate.

10. Launch Acceptance Criteria

- Mobile responsiveness is verified on iOS Safari and Android Chrome across 360–430 px viewports.
- Saree photo upload displays an interactive magnifying loupe without material frame stutter or touch latency.
- The eyedropper converts sampled pixels into CIELAB coordinates and triggers perceptual catalog matching.
- Admin product creation supports image upload, color sampling, Hex storage, and correct CIELAB persistence.
- AI Studio validates the 800 × 800 px minimum and rejects blurry, unsupported, or oversized files.
- The asynchronous generation pipeline returns front, back, and 3/4 views within 45 seconds under the agreed normal operating load.

- ■ AI renders maintain consistent fabric color and pattern across all three generated views within an acceptable visual tolerance.
- ■ Public /d/:id pages resolve without authentication and display the completed views and style tags.
- ■ Order via WhatsApp produces a valid URI-encoded enquiry and opens WhatsApp Business.
- ■ When V1 commerce is disabled, cart, payment, checkout, and shipping UI are not exposed.

11. Non-Functional Requirements

- **Performance:** Catalog and product pages should remain lightweight and responsive on mobile networks.
- **Reliability:** AI generation failures must be captured with a user-safe error state and an auditable provider status.
- **Security:** User uploads and administrative functions must be protected by appropriate authentication and authorization controls.
- **Privacy:** Public design pages should expose only assets and metadata intended for sharing; private user profile data must not be public.
- **Scalability:** AI work must remain decoupled from the web request lifecycle so provider latency does not block the application.
- **Observability:** Generation status, latency, failure rate, provider job identifiers, and quota usage should be measurable.

12. V1 Boundaries & Deferred Capabilities

Capability	V1	V2
Product catalog & discovery	Included	Retain
Saree color matching	Included	Retain / improve
AI blouse generation	Included	Retain / improve
Public design sharing	Included	Retain
WhatsApp assisted order	Included	Optional fallback
Online cart & checkout	Excluded	Enable
Online payments	Excluded	Enable
Atomic inventory reservation	Excluded	Enable
Automated shipping / AWB	Excluded	Enable
3D / 360° viewer	Excluded	Evaluate / enable

13. Implementation Readiness Summary

V1 is considered implementation-ready when the engineering team can build the catalog, color matching, AI Studio, public design sharing, and WhatsApp conversion flow without requiring the deferred V2 commerce systems. All acceptance criteria in Section 10 must be testable independently, and feature flags must provide a clean boundary between assisted ordering and future automated commerce.